Jiatao **Yan**

AI Postgraduate Student with a Focus on Computer Vision & Medical AI

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Education

Sussex Artificial Intelligence Institute, Zhejiang Gongshang University — *Hangzhou, China*

Master in Artificial Intelligence and Adaptive System — *Sept 2024 – Jan 2026 (Expected)*

- Thesis: Integrating physics models into VAE framework to enhance latent space interpretability.
- Core Courses: Image Processing, NLP, ML, Intelligence in Animals and Machines, Python Programming.

Wenzhou Business College — *Wenzhou, China*

B.S. in Computer Science and Technology — *Sept 2019 – June 2023*

- GPA: 3.41/5.0 (84.7/100)
- Thesis: Smoking Behavior Detection Based on Deep Learning and Skeletal Frameworks.

Research & Internship Experience

First Affiliated Hospital of Wenzhou Medical University - Hepatobiliary and Pancreatic Surgery Laboratory — *Wenzhou, China*

Research Intern — *Sept 2022 – Jan 2023*

- Developed medical image preprocessing software, obtained software copyright (2022SR0252378).
- Contributed to deep learning model for leukemia diagnosis and exosome feature analysis in hepatocellular carcinoma research.

Zhejiang University Urban Planning and Design Institute — *Hangzhou, China*

Research Intern — *Sept 2025 – Dec 2025*

- Developed intelligent document processing and multi-agent workflow systems for government/enterprise scenarios.
- Introduced MCP tools and pluggable toolchains, implemented multi-agent collaboration and observability.

Huawei Ascend NPU Migration Project — *Hangzhou, China*

Government-Enterprise Project Participant — *Sept 2025 – Dec 2025*

- Migrated CUDA-based YOLOv8 training/inference pipeline to 8×Huawei Ascend 910B (NPU).
- Completed CANN/ACL adaptation and HCCL multi-card training, fixed critical operator differences and aligned precision.
- Optimized data pipeline and graph mode execution for containerized deployment while maintaining mAP performance.

National Innovation Training Program for University Students (Project No. 202213637004) — *Wenzhou, China*

Project Lead — *June 2022 – June 2023*

- First principal investigator of national-level research project. Responsible for technical design, team management, and project completion.

Publications

Machine Learning Identifies Exosome Features Related to Hepatocellular Carcinoma

Published in *Frontiers in Cell and Developmental Biology* — *Sept 2022*

- Authors: Kai Zhu, Qiqi Tao, ***Jiatao Yan***, Zhichao Lang, Xinmiao Li, Yifei Li, Congcong Fan, Zhengping Yu
- DOI: 10.3389/fcell.2022.1020415
- Contribution (Co-first author, 3rd): Designed ML pipeline, applied Random Forest/SVM-RFE/LASSO algorithms to identify key biomarkers from exosome proteomics data.

Multi-omics and Machine Learning-driven CD8+ T Cell Heterogeneity Score for Prognosis

Published in *Molecular Therapy Nucleic Acids* — *Dec 2024*

- Authors: Di He, Zhan Yang, Tian Zhang, Yaxian Luo, Lianjie Peng, ***Jiatao Yan***, et al.
- DOI: 10.1016/j.omtn.2024.102413
- Contribution: Implemented LASSO regression and other ML algorithms to identify key genes for HNSCC prognosis, supporting CD8+ T cell heterogeneity score construction.

Using Multiomics and Machine Learning: Insights into Improving the Outcomes of Clear Cell Renal Cell Carcinoma via the SRD5A3-AS1/hsa-let-7e-5p/RRM2 Axis

Published in *ACS Omega* — *June 2025*

- Authors: Mouyuan Sun, Zhan Yang, Yaxian Luo, Luying Qin, Lianjie Peng, Chaoran Pan, ***Jiatao Yan***, Tao Qiu, Yan Zhang
- DOI: 10.1021/acsomega.5c01337
- Contribution: Implemented ML algorithms to validate SRD5A3-AS1/hsa-let-7e-5p/RRM2 signaling axis role in clear cell renal cell carcinoma.

A multi-data fusion deep learning model for prognostic prediction in upper tract urothelial carcinoma

PUBLISHED IN *Frontiers in Oncology* *Aug 2025*

- Authors: Hongdi Sun, Siping Chen, Yongxing Bao, Fengyan You, Honghui Zhu, Xin Yao, Lianguo Chen, Jiangwei Miao, Fanggui Shao, Xiaomin Gao, Binwei Lin
- DOI: 10.3389/fonc.2025.1644250
- Contribution: Designed DL architecture for multi-phase CT analysis, developed methods to integrate imaging features with clinical data for survival prediction.

Manuscripts Under Review & In Preparation

YOLOv11-LCDFS: Enhanced Smoking Detection With Low-light Enhancement

Under Review

- Authors: ***Jiatao Yan***, Zhuzikai Zheng, Zhengtan Yang, Hao Jiang, Peichen Wang, Fangjun Kuang, Siyang Zhang
- Contribution (First author): Developed YOLO-based architecture with low-light enhancement, attention mechanisms, and optimized upsampling.

A Spatio-Temporal Graph Transformer for Decoding Motor Imagery from fNIRS in Post-Stroke Patients

In Preparation

- Authors: ***Jiatao Yan*** (First author)
- Contribution: Proposed STGT for fNIRS motor imagery decoding, achieved 79.8% LOSO accuracy, outperforming SVM/CNN/LSTM baselines.

Validating TCM Kidney–Brain Co-Treatment Theory for Post-Stroke Dysphagia via Explainable Multimodal AI

In Preparation

- Authors: ***Jiatao Yan*** (First author)
- Contribution: Fused 3D brain lesion imaging with clinical features, validated "kidney→brain→swallowing" causal chain via mediation testing.

A Deep Learning Model for Automated Sonographic Assessment of Diastasis Recti Abdominis

In Preparation

- Authors: ***Jiatao Yan*** (First author)
- Contribution: Proposed DL model for automated ultrasound assessment of DRA, completed rectus abdominis segmentation and geometric point estimation.

Projects & Achievements

2024	**Bronze Medal, Ranked 143rd/1516 (Top 10%)**, Stanford RNA 3D Folding	*Kaggle Competition*
2024	**Ranked 116th/3310 (Top 4%)**, Predict Podcast Listening Time	*Kaggle Competition*
2024	**Ranked 178th/4316 (Top 5%)**, Predict Calorie Expenditure	*Kaggle Competition*
May 2023	**Bronze Medal**, 18th "Challenge Cup" Provincial College Student Competition	*Zhejiang Province*
Apr 2024	**Student Member Computer Innovation and Entrepreneurship Award**, Wenzhou Computer Society	*Wenzhou, China*

Patents & Intellectual Property

Invention Patent, Smoking Behavior Recognition Camera and Method	*Application No. 202310277784.1*
Software Copyright, Medical Image Computing Software	*2022SR0252378*
Software Copyright, Human Skeleton Recognition Software	*2022SR1258998*
Software Copyright, Cigarette Recognition Software	*2022SR1277520*
Software Copyright, Smoking Behavior Detection Software	*2022SR1277521*

Technical Skills

Programming	Python, C, Java, SQL, C#, JavaScript, Vue
ML/DL	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy
Tools	Git, Docker, LaTeX, GeoPandas, Matplotlib, Seaborn
Languages	Mandarin (Native), English (IELTS 6.0, CET6)